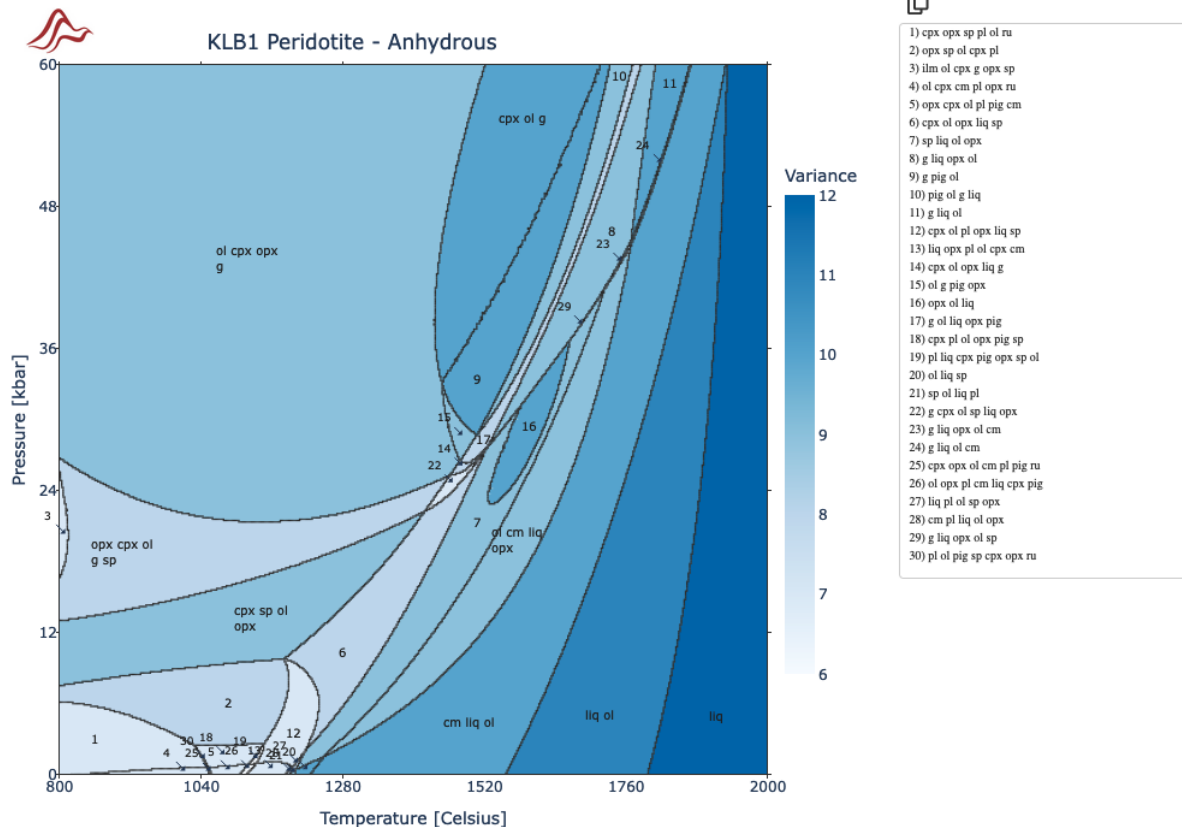


MAGEMinApp: introduction - v0.6.2



Main contributors

Alexandre Peillod
Anton Popov
Boris Kaus
Hugo Dominguez
Hendrik Ranocha
Jean-François Moyen
Joshua Laughton
Jun Ren
Lorenzo Candioti
Owen Weller
Pierre Lanari
Simon Scholz
Tim J.B. Holland
Tobias Keller



MAGEMin github: ComputationalThermodynamics

- First, download the resources for the shortcourse from Github:



<https://github.com/ComputationalThermodynamics/Resources>

Resources (Public)

main 1 Branch 0 Tags

Go to file Add file Code

Files

- GG2024 corrected typos
- Mainz2025 adding Mainz25 sh
- LICENSE Initial commit
- README.md adding Mainz25 sh

Clone

Local Codespaces Copilot

HTTPS SSH GitHub CLI

<https://github.com/ComputationalThermodynamar>

Clone using the web URL.

Open with GitHub Desktop

Download ZIP

About

No description, website, or topics provided.

Readme

GPL-3.0 license

Activity

Custom properties

2 stars

3 watching

0 forks

Report repository

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Languages

Julia 100.0%

Resources

This repository stores various resources and support materials used during MAGEMin shortcourses and workshops. It provides examples on how to use [MAGEMin](#), [MAGEMin_C](#) and [MAGEMinApp](#).

- Mainz2025 - MAGEMinApp introduction shortcourse in Mainz, Germany (Jan, 7, 15, 2025)
 - Set of tutorials on how to use [MAGEMinApp](#)
 - Used versions: MAGEMin v1.6.2, MAGEMin_C v1.6.9 and MAGEMinApp v0.6.2
- GG2024 - MAGEMin workshop at Granulites & Granulites 2024 conference in Verbania, Italy (Sep 3-6, 2024)
 - Set of tutorials on how to use [MAGEMinApp](#) and [MAGEMin_C](#)

- The Resources folders include:

- Bulk rock input file example
- Trace element input file
- Reference database used in MAGEMinApp
- The pdfs used in the short course
- MAGEMin_C Julia scripts examples (not treated here)

MAGEMinApp: launching the App in parallel

VISUAL STUDIO CODE

MAC, LINUX AND WINDOWS(USING WSL)

```
seph42@DESKTOP-2V82075:~$ julia -t 6

Documentation: https://docs.julialang.org
Type "?" for help, "]" for Pkg help.

Version 1.10.0 (2023-12-25)
Official https://julialang.org/ release

julia> using MAGEMinApp
App()
Using libMAGEMin.dylib from MAGEMin_jll

julia> App()
[ Info: Listening on: 127.0.0.1:8050, thread id: 2
```

```
julia -t 6
using MAGEMinApp
App()
```

Favored way
(better performances on
Mac and Linux)

POWERSHELL

WINDOWS(NO WSL)

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\Seph> $env:JULIA_NUM_THREADS=6
PS C:\Users\Seph> julia

Documentation: https://docs.julialang.org
Type "?" for help, "]" for Pkg help.

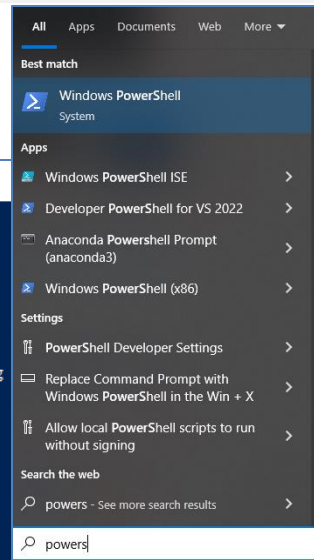
Version 1.9.3 (2023-08-24)
Official https://julialang.org/ release

julia> using MAGEMinApp
Using libMAGEMin.dylib from MAGEMin_jll

julia> App()
[ Info: Listening on: 127.0.0.1:8050, thread id: 1
```

```
$env:JULIA_NUM_THREADS=6
julia
using MAGEMinApp
App()
```

Around 2.5 times slower
than using WSL



MAGEMinApp: interface (Phase diagrams)



Reminder: Don't forget to properly cite the references used to create the diagrams by using the 'export references' buttons!



Phase diagrams

PTX path

Isentropic path

General information

Setup

Diagram

Trace-elements

Phase diagram parameters

Thermodynamic database

Igneous (Green et al., 2025, after H18)

Solution phase selection

Diagram type

P-T diagram

H₂O-saturated at solidus

false

☐ Limit Ca-opx

0.5

TE predictive model

false

Pressure [kbar]

min

0.01

max

20.01

Temperature [°C]

min

800

max

1400

Initial grid subdivision

3

8 × 8 grid

Refinement type

Phases only

Refinement levels

2

Buffer

no buffer

Solver

Hybrid

Verbose

none

Specific Cp

GO

Bulk-rock composition

mol%

Drag and drop or select bulk-rock file

KLB1 Peridotite - Anhydrous

oxide	fraction
SiO2	38.494
Al2O3	1.776
CaO	2.824
MgO	50.566
FeO	5.886
K2O	0.01
Na2O	0.25
TiO2	0.1
O	0.096
Cr2O3	0.109
H2O	0

ig

General parameters

Title

KLB1 Peridotite - Anhydrous

Update

Reset

Compute phase diagram

Save/Load Diagram

filename

Save state

Load state

State/CSV directory: /Users/nicolazic/julia/packages/MAGEMinApp/rIEFD

Help and contact

Links

[Tutorials](#)
[Post issue](#)
[Open discussion](#)

Comments

Set of example on how to use MAGEMinApp
Something is not working properly?
Need additional options?

Contributors

MAGEMinApp: interface (Phase diagrams)

Progress bar

Phase diagrams **PTX path** Isentropic path General information
Setup **Diagram** Trace-elements

Phase diagram parameters

Thermodynamic database: Igneous (Green et al., 2025, after H18) ▼

Solution phase selection: ☐

Diagram type: P-T diagram ▼

H₂O-saturated at solidus: false ▼

☐ Limit Ca-opx: 0.5 ▲▼

TE predictive model: false ▼

Pressure [kbar]: min 0.01 ▲▼ max 20.01 ▲▼

Temperature [°C]: min 800 ▲▼ max 1400 ▲▼

Initial grid subdivision: 3 ▲▼ 8 x 8 grid

Refinement type: Phases only ▼

Refinement levels: 2 ▲▼

Buffer: no buffer ▼

Solver: Hybrid ▼

Verbose: none ▼

Grid properties

Bulk-rock composition

mol% ▼

Drag and drop or select bulk-rock file

KLB1 Peridotite - Anhydrous ▼

oxide	fraction
SiO2	38.494
Al2O3	1.776
CaO	2.824
MgO	50.566
FeO	5.886
K2O	0.01
Na2O	0.25
TiO2	0.1
O	0.096
Cr2O3	0.109
H2O	0

Bulk rock composition(s)

General parameters

Title: KLB1 Peridotite - Anhydrous

Update Reset

Compute phase diagram

Save/Load Diagram

filename Save state Load state

State/CSV directory: /Users/nicolasriell/julia/packages/MAGEMinApp/rHEPD

Help and contact

Links
[Tutorials](#)
[Post issue](#)
[Open discussion](#)

Comments
Set of example on how to use MAGEMinApp
Something is not working properly?
Need additional options?

- Save/load state
- Compute diagram

MAGEMinApp: Thermodynamic database

Phase diagrams | **PTX path** | Isentropic path | General information

Setup | **Diagram** | Trace-elements

Phase diagram parameters

Thermodynamic database

Solution phase selection

Diagram type

H₂O-saturated at solidus

TE predictive model

Pressure [kbar]

Temperature [°C]

Initial grid subdivision

Refinement type

Refinement levels

Buffer

Solver

Verbose

Specific Cp

Igneous (Green et al., 2025, after H18)

- PUBLISHED DATABASE -

Metapelite (White et al., 2014)

Metabasite (Green et al., 2016)

Igneous (Green et al., 2025, after H18)

Igneous alkaline dry (Weller et al., 2024)

Ultramafic (Evans & Frost., 2021)

min max

0.01 20.01

min max

800 1400

3 8 × 8 grid

Phases only

2

no buffer

Hybrid

none

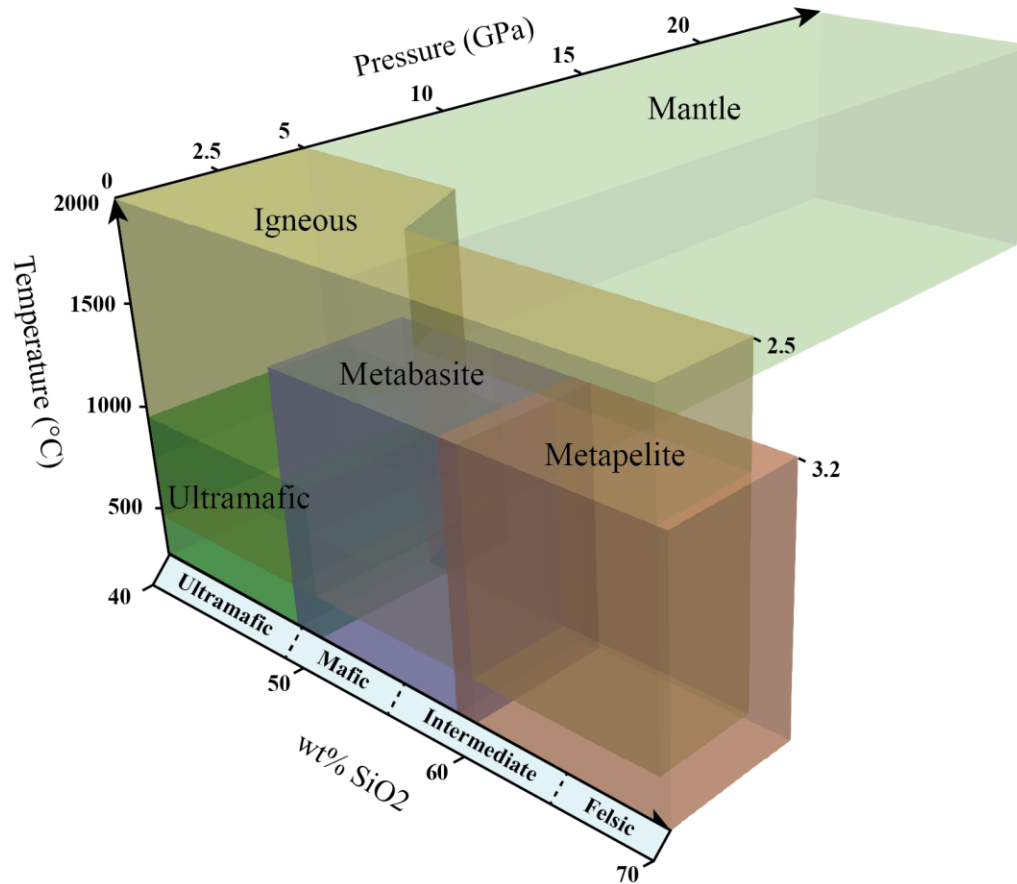
G0

id=ee5b2c93-cc64-4fa1-8fc2-2eb01402fb0f; MAGEMinApp GUI v=0.6.2; using 12/12 threads

- **Mantle (Holland et al., 2013)**
NCFMAS
- **Metapelite (White et al., 2014)**
MnNCKFMASHTO
- **Metabasite (Green et al., 2016)**
NCKFMASHTO
- **Igneous (Green et al., 2025 updated after Holland et al., 2018)**
NCKFMASHTOCr
- **Ultramafic (Evans & Frost, 2021)**
FMASOSH
- **Dry igneous alkaline (Weller et al., 2024)**
NCKFMASHTOCr
- **Mantle (Stixrude and Lithgow-Bertelloni, 2011)**
NCFMAS (upcoming)

MAGEMinApp: Thermodynamic database

- Which database for (roughly) which conditions?



MAGEMinApp: solution phase selection

Phase diagram parameters

Thermodynamic database: Igneous (Green et al., 2025, after H18) ▼

Solution phase selection

- ☒ spn
- ☒ bi
- ☒ cd
- ☒ cpx
- ☒ ep
- ☒ g
- ☒ hb
- ☒ ilm
- ☒ liq
- ☒ ol
- ☒ opx
- ☒ fsp
- ☒ fl
- ☒ mu
- ☒ fper

Diagram type: P-T diagram ▼

H₂O-saturated at solidus: false ▼

☐ Limit Ca-opx 0.5

- Allow to deactivate solution phase models
- Note that solution phase models from different dataset cannot be mixed (yet)

MAGeMinApp: diagram types

Phase diagram parameters

Thermodynamic database

Igneous (Green et al., 2025, after H18) ▼

Solution phase selection

☐

Diagram type

P-T diagram ▲

P-T diagram

P-X diagram

T-X diagram

PT-X diagram

TE predictive model

Pressure [kbar]

min
0.01 ▲▼

max
20.01 ▲▼

Temperature [°C]

min
800 ▲▼

max
1400 ▲▼

Initial grid subdivision

3 ▲▼

8 × 8 grid

Refinement type

Phases only ▼

Refinement levels

2 ▲▼

Buffer

no buffer ▼

Solver

Hybrid ▼

Verbose

none ▼

Specific Cp

G0 ▼

P-T

Pressure [kbar]
Temperature [°C]

P-X

Fixed
temperature

Pressure [kbar]
Composition
[bulk1-bulk2]

T-X

Fixed
pressure

Temperature [°C]
Composition
[bulk1-bulk2]

PT-X

PT path
Composition
[bulk1-bulk2]

MAGEMinApp: Trace element predictive model

Implemented with the help of J-F Moyen (LMV) and H. Dominguez (Uni - Bern)

Phase diagram parameters

Thermodynamic database: Igneous (Green et al., 2025, after H18)

Solution phase selection: ☐

Diagram type: P-T diagram

H₂O-saturated at solidus: false

☐ Limit Ca-opx: 0.5

TE predictive model: true

Kd's database: true

Zr saturation: false

Pressure [kbar]: 0.01 (min) to 20.01 (max)

Temperature [°C]: 800 to 1400

Initial grid subdivision: 3 (min) to 8 x 8 grid (max)

Refinement type: Phases only

Refinement levels: 2

Buffer: no buffer

Solver: Hybrid

Verbose: none

Specific Cp: G0

- Activate trace-element partitioning at supra-solidus conditions
- Displays trace element bulk panel on the right
- Gives the option to select among 3 zirconium saturation models

Watson & Harrison (1983)

Boehnke et al. (2013)

Crisp and Berry (2022)

General parameters

Title: KLB1 Peridotite - Anhydrous

Update Reset

filename Save state Load state

Compute phase diagram

State/CSV directory: /home/seph42/MAGEMinApp/jl_v04.0

Trace-element composition

Drag and drop or select trace-element file

Pyrolite (Primitive Mantle)

elements	µg/g
Rb	0.6
Ba	6.6
Th	0.0795
U	0.0203
Nb	0.658
Ta	0.037
La	0.648
Ce	1.675
Pb	0.15
Pr	0.254
Sr	19.9
Nd	1.25

MAGEMinApp: grid subdivision and refinement

Phase diagram parameters

Thermodynamic database: Igneous (Green et al., 2025, after H18) ▼

Solution phase selection: ☐

Diagram type: P-T diagram ▼

H₂O-saturated at solidus: false ▼

☐ Limit Ca-opx: 0.5 ▼

TE predictive model: false ▼

Pressure [kbar]: min 0.01 ▼ max 20.01 ▼

Temperature [°C]: min 800 ▼ max 1400 ▼

Initial grid subdivision: 3 ▼ 8 × 8 grid

Refinement type: Phases only ▼

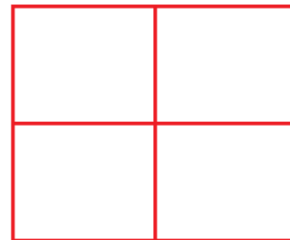
Refinement levels: 2 ▼

Buffer: no buffer ▼

Solver: Hybrid ▼

Verbose: none ▼

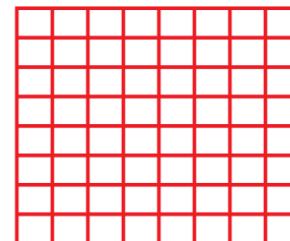
Specific Cp: G0 ▼



**1 initial grid
subdivision**



**2 initial grid
subdivisions**



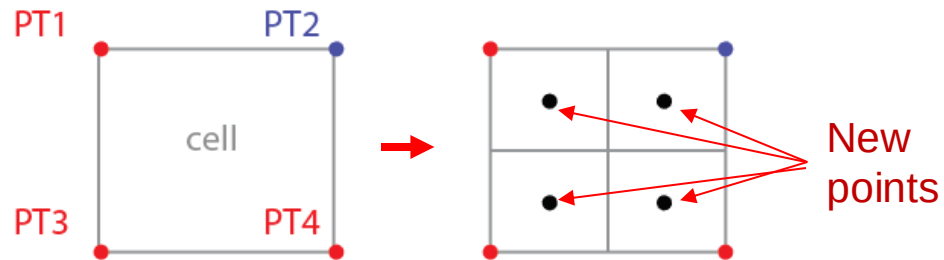
**3 initial grid
subdivisions**

MAGEMinApp: grid subdivision and refinement

Phase diagram parameters

Thermodynamic database	Igneous (Green et al., 2025, after H18)
Solution phase selection	☐
Diagram type	P-T diagram
H ₂ O-saturated at solidus	false
☐ Limit Ca-opx	0.5
TE predictive model	false
Pressure [kbar]	0.01 min 20.01 max
Temperature [°C]	800 min 1400 max
Initial grid subdivision	3 8 × 8 grid
Refinement type	Phases only
Refinement levels	2
Buffer	no buffer
Solver	Hybrid
Verbose	none
Specific Cp	G0

- **Phase only**
refines (split) cells when a phase change is recorded in one of the four corners
- **Dominant end-member**
refines (split) cells when the dominant end-member of any solution phase model changes in one of the four corners



- **Refinement levels**
Number of successive grid refinements

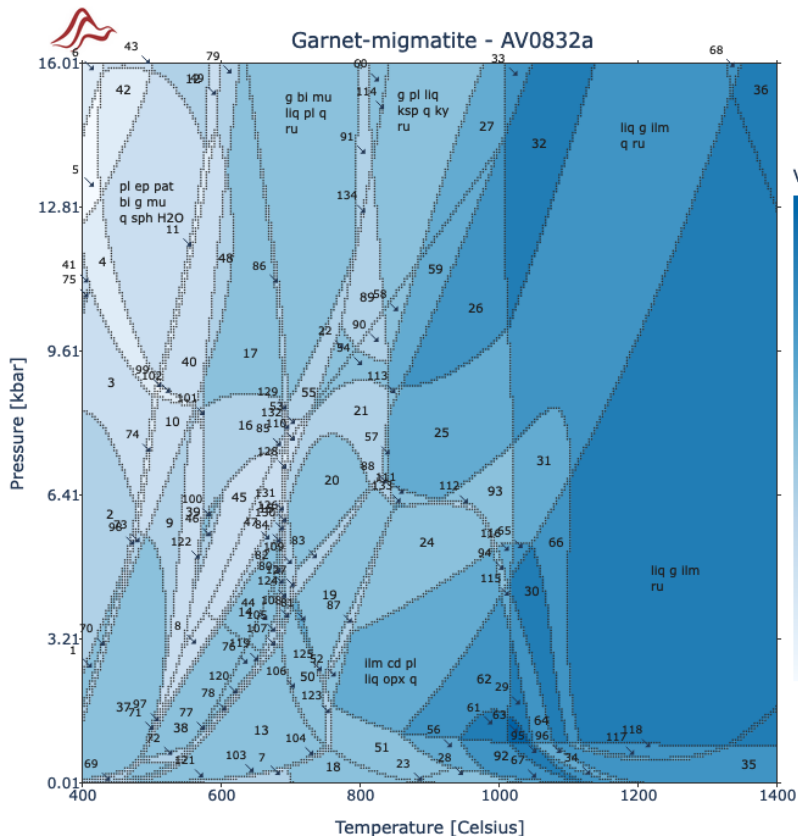
MAGEMinApp: grid refinement

Export ϕ for LaMEM Export for GeoModel



Refine uniformly

Refine phase boundaries



- 1) mu bi chl pl q sph H2O
- 2) chl ep bi mu pl q sph H2O
- 3) mu chl ep bi pl g q sph H2O
- 4) pat ep chl bi pl mu g q sph H2O
- 5) pat bi mu chl g pl ep opx q sph H2O
- 6) pat g mu opx bi pl chl q sph H2O
- 7) opx pl ksp ilm sp cd q H2O
- 8) pl bi ilm mu g q and ru H2O
- 9) bi mu g chl pl q ru H2O
- 10) chl pl bi ep g mu q ru H2O
- 11) pat bi g mu pl ep q ru sph H2O
- 12) pat mu g bi pl q sph H2O
- 13) cd ilm pl ksp bi q H2O
- 14) bi pl mu q sill ru H2O
- 15) ilm bi g pl mu q sill ru H2O
- 16) bi pl g ilm mu q ru H2O
- 17) g bi mu pl q ru H2O
- 18) opx pl cd ilm ksp q H2O
- 19) pl ilm cd bi g liq q
- 20) bi ilm pl g liq q sill
- 21) bi ilm g pl liq q sill ru
- 22) g pl liq bi q ky ru
- 23) cd ilm ksp pl opx trd H2O
- 24) g liq ilm pl opx cd q
- 25) g ilm liq pl q ru
- 26) pl g liq q ru
- 27) g liq pl q ky ru
- 28) cd pl ilm liq opx trd H2O
- 29) liq ilm cd opx
- 30) opx liq ilm ru
- 31) g liq ilm opx q ru
- 32) g liq q ru
- 33) g liq q ky ru
- 34) liq ilm cd ru H2O
- 35) liq ilm opx ru H2O
- 36) hem liq g ru
- 37) pl chl mu bi q ru H2O
- 38) bi pl mu bi q and ru H2O
- 39) g pl mu bi chl ilm q ru H2O
- 40) ep g bi pl pat mu q ru H2O
- 41) pat ep chl mu pl g q sph H2O
- 42) pat mu g bi pl ep opx q sph H2O
- 43) pat bi mu opx pl e a sph H2O

Informations

Display options

Isopleths

Classifications

Select field to display

Field

Variance

Diagram options

Show reaction lines

true

Show stable phases

true

Show grid

false

Color options

Colormap

Blues

Value range

min

2

max

10

Colormap range



Set min to white

false

Reverse colormap

false

Smooth colormap

best

MAGEMinApp: grid refinement

Export ϕ for LaMEM Export for GeoModel



Refine uniformly

Refine phase boundaries



Informations

Display options

Isolepts

Classifications

Select field to display

Field

Variance

Diagram options

Show reaction lines

true

Show stable phases

true

Show grid

true

Color options

Colormap

Blues

min

max

Value range

2

10

Colormap range

1 2 3 4 5 6 7 8 9

Set min to white

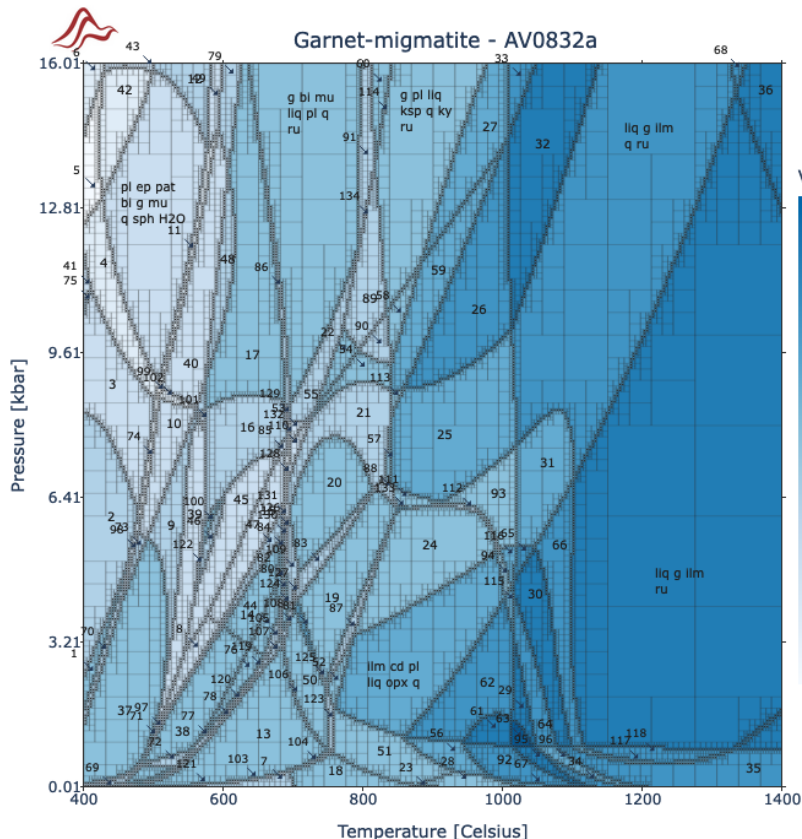
false

Reverse colormap

false

Smooth colormap

best



Variance

- 1) mu bi chl pl q sph H2O
- 2) chl ep bi mu pl q sph H2O
- 3) mu chl ep bi pl g q sph H2O
- 4) pat ep chl bi pl mu g q sph H2O
- 5) pat bi mu chl g pl ep opx q sph H2O
- 6) pat g mu opx bi pl chl q sph H2O
- 7) opx pl ksp ilm sp cd q H2O
- 8) pl bi ilm mu g q and ru H2O
- 9) bi mu g chl pl q ru H2O
- 10) chl pl bi ep g mu q ru H2O
- 11) pat bi g mu pl ep q ru sph H2O
- 12) pat mu g bi pl q sph H2O
- 13) cd ilm pl ksp bi q H2O
- 14) bi pl mu q sill ru H2O
- 15) ilm bi g pl mu q sill ru H2O
- 16) bi pl g ilm mu q ru H2O
- 17) g bi mu pl q ru H2O
- 18) opx pl cd ilm ksp q H2O
- 19) pl ilm cd bi g liq q
- 20) bi ilm pl g liq q sill
- 21) bi ilm g pl liq q sill ru
- 22) g pl liq bi q ky ru
- 23) cd ilm ksp pl opx trd H2O
- 24) g liq ilm pl opx cd q
- 25) g ilm liq pl q ru
- 26) pl g liq q ru
- 27) g liq pl q ky ru
- 28) cd pl ilm liq opx trd H2O
- 29) liq ilm cd opx
- 30) opx liq ilm ru
- 31) g liq ilm opx q ru
- 32) g liq q ru
- 33) g liq q ky ru
- 34) liq ilm cd ru H2O
- 35) liq ilm opx ru H2O
- 36) hem liq g ru
- 37) pl chl mu bi q ru H2O
- 38) bi pl mu q and ru H2O
- 39) g pl mu bi chl ilm q ru H2O
- 40) ep g bi pl pat mu q ru H2O
- 41) pat ep chl mu pl g q sph H2O
- 42) pat mu g bi pl ep opx q sph H2O
- 43) pat bi mu opx pl g q sph H2O

MAGEMinApp: buffers

Phase diagram parameters

Thermodynamic database: Igneous (Green et al., 2025, after H18) ▼

Solution phase selection: ☐

Diagram type: P-T diagram ▼

H₂O-saturated at solidus: false ▼

☐ Limit Ca-opx: 0.5 ▼

TE predictive model: false ▼

Pressure [kbar]: min 0.01 ▼ max 20.01 ▼

Temperature [°C]: min 800 ▼ max 1400 ▼

Initial grid subdivision: 3 ▼ 8 × 8 grid

Refinement type: Phases only ▼

Refinement levels: 2 ▼

Buffer: no buffer ▼

Solver: Hybrid ▼

Verbose: none ▼

Specific Cp: G0 ▼

- **Oxygen buffers**
QFM, MW, QIF, CCO, HM, NNO
- **Fix system activity**
H₂O, O₂, SiO₂, FeO, MgO, Al₂O₃, TiO₂

Bulk-rock composition

mol% ▼

Drag and drop or select bulk-rock file

KLB1 Peridotite - Anhydrous ▼

oxide	fraction
SiO ₂	38.494
Al ₂ O ₃	1.776
CaO	2.824
MgO	50.566
FeO	5.886
K ₂ O	0.01
Na ₂ O	0.25
TiO ₂	0.1
O	0.096
Cr ₂ O ₃	0.109
H ₂ O	0

Buffer multiplier: 0 ▼

Igneous (Green et al., 2025, after H18)

- **Selecting a buffer displays the option to apply an offset in RTlog scale**
e.g., +2 QFM

- **Allow to provide the activity value in case activity is selected**

MAGEMinApp: solvers

Phase diagram parameters

Thermodynamic database	Igncous (Green et al., 2025, after H18) ▼	
Solution phase selection	☐	
Diagram type	P-T diagram ▼	
H ₂ O-saturated at solidus	false ▼	
☐ Limit Ca-opx	0.5 ▼	
TE predictive model	false ▼	
Pressure [kbar]	min 0.01 ▼	max 20.01 ▼
Temperature [°C]	min 800 ▼	max 1400 ▼
Initial grid subdivision	3 ▼ 8 × 8 grid	
Refinement type	Phases only ▼	
Refinement levels	2 ▼	
Buffer	no buffer ▼	
Solver	Hybrid ▼	
Verbose	none ▼	
Specific Cp	G0 ▼	

- **Default MAGEMin solver to perform stable phase equilibria calculation**
No need to be changed in normal use

MAGEMinApp: solvers

Phase diagram parameters

Thermodynamic database	Igneous (Green et al., 2025, after H18) ▼	
Solution phase selection	☐	
Diagram type	P-T diagram ▼	
H ₂ O-saturated at solidus	false ▼	
☐ Limit Ca-opx	0.5 ▼	
TE predictive model	false ▼	
Pressure [kbar]	min 0.01 ▼	max 20.01 ▼
Temperature [°C]	min 800 ▼	max 1400 ▼
Initial grid subdivision	3 ▼	8 × 8 grid
Refinement type	Phases only ▼	
Refinement levels	2 ▼	
Buffer	no buffer ▼	
Solver	Hybrid ▼	
Verbose	none ▼	
Specific Cp	G0 ▼	

- **Displays details of the calculation(s) in the Julia terminal**
mostly used for debugging

MAGEMinApp: Specific heat capacity

Phase diagram parameters

Thermodynamic database	Igneous (Green et al., 2025, after H18) ▼	
Solution phase selection	☐	
Diagram type	P-T diagram ▼	
H ₂ O-saturated at solidus	false ▼	
☐ Limit Ca-opx	0.5 ▼	
TE predictive model	false ▼	
Pressure [kbar]	min 0.01 ▼	max 20.01 ▼
Temperature [°C]	min 800 ▼	max 1400 ▼
Initial grid subdivision	3 ▼	8 × 8 grid
Refinement type	Phases only ▼	
Refinement levels	2 ▼	
Buffer	no buffer ▼	
Solver	Hybrid ▼	
Verbose	none ▼	
Specific Cp	G0 ▼	

- Method to compute specific heat capacity

$$C_p = -T \frac{\Delta^2 G}{\Delta T^2} = -T \frac{G^{(T+\delta T)} + G^{(T-\delta T)} - 2G^{(T)}}{\delta T^2}$$

where C_p is the specific heat capacity, T is the temperature

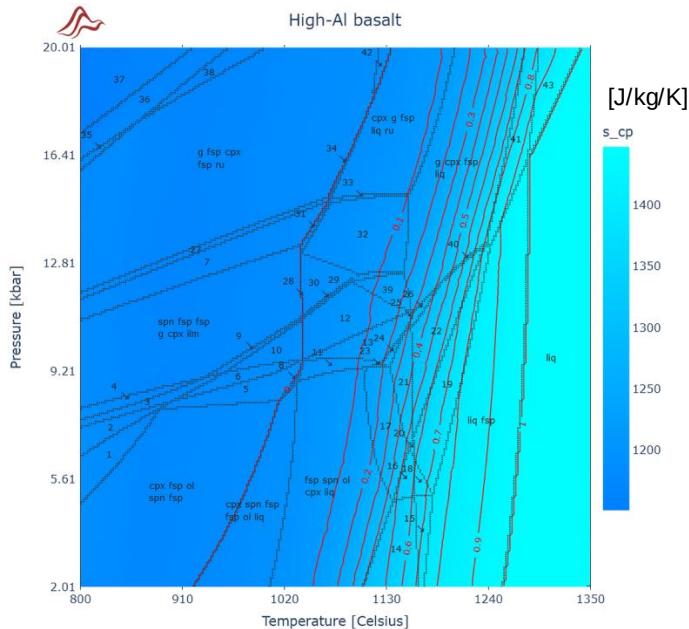
- Can be computed using

1. phase aggregate (G_0) **Without reactions**
2. phase assemblage (G_{system}) **With reactions**

- Note that when **G_system** is selected, solver must be set to legacy
higher accuracy at suprasolidus conditions is needed to compute Gibbs second derivative

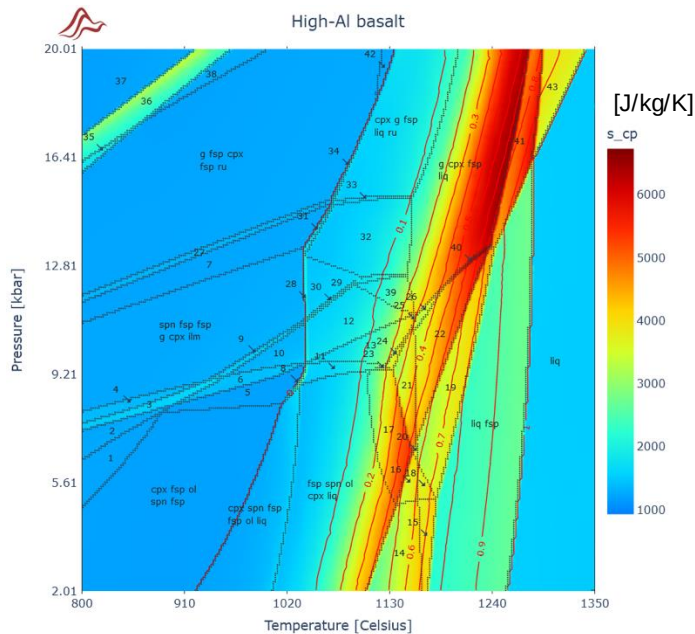
Specific heat capacity: what difference does it make?

G_0 differentiation (Cp without reaction)



- Cheap to compute
- Ignores latent heat

G_{system} differentiation (Cp with reaction)



- More expensive to compute
- Correct heat budget
- To be chosen for geodynamic coupling

MAGEMinApp: bulk rock composition

Bulk-rock composition

mol% ▼

Drag and drop or select bulk-rock file

KLB1 Peridotite - Anhydrous ▼

oxide	fraction
SiO ₂	38.494
Al ₂ O ₃	1.776
CaO	2.824
MgO	50.566
FeO	5.886
K ₂ O	0.01
Na ₂ O	0.25
TiO ₂	0.1
O	0.096
Cr ₂ O ₃	0.109
H ₂ O	0

Buffer multiplier 0

Igneous (Green et al., 2025, after H18)

- Change system unit of displayed composition mol, wt
- Load custom bulk rock composition(s)
- Select predefined bulk rock composition for testing
- Bulk rock composition displayed in mol%
note that only the oxides taken into account in the database are displayed

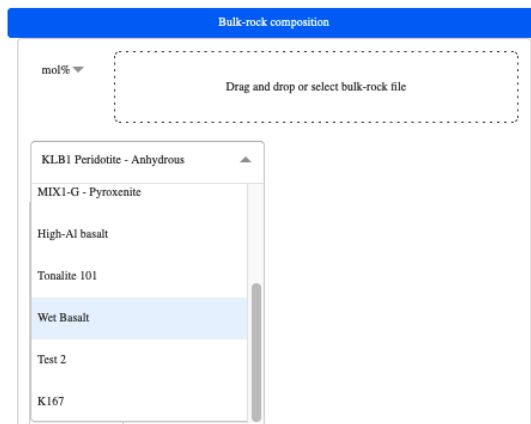
MAGEMinApp: bulk rock input file format

Example of input file provided in:

https://github.com/ComputationalThermodynamics/Resources/Mainz2025/bulkrock_composition_input_files/

```
# HEADER
title; comments; db; sysUnit; oxide; frac; frac2
# BULK-ROCK-COMPOSITION
Test 2; Moo-et-al., 2000; ig; mol; [SiO2, Al2O3, CaO, MgO, FeO, K2O, Na2O, TiO2, O, Cr2O3, H2O]; [48.97, 11.76, 13.87, 4.21, 8.97, 1.66, 10.66, 1.36, 1.66, 0.01, 5.0];
K167; Moo-et-al., 2000; ig; mol; [SiO2, Al2O3, CaO, MgO, FeO, K2O, Na2O, TiO2, O, Cr2O3, H2O]; [50.3162, 8.0778, 11.6616, 11.3262, 9.1893, 0.4312, 2.4051, 2.271, 4.312, 0.0096, 0.0];
Test 3; Coin & Kwak, 1984; mb; wt; [SiO2, Al2O3, CaO, MgO, Fe2O3, K2O, Na2O, TiO2, FeO, H2O]; [55.12, 12.76, 4.32, 5.21, 2.45, 1.66, 10.66, 1.36, 1.66, 2.0];
Simon; Shorn-Simon, 2023; mb; mol; [SiO2, Al2O3, CaO, MgO, FeO, K2O, Na2O, TiO2, O, H2O]; [61.8, 8.57, 9.26, 5.05, 7.27, 0.21, 3.43, 1.25, 0.73, 2.85];
```

- Note that either [..., Fe2O3, FeO,...] or [..., FeO, O, ...] must be provided for the database accounting for O



Once loaded, the bulk-rock composition appear in the dropdown menu

Note that the bulk are attached to a database, this is why only Test 2 and K167 appear in the dropdown menu